MUHAMMAD USAMA BIN AKHTAR KHAN

Applied AI Engineer

Computer Vision · Local GPU Inference · Full-Stack (React/Next.js, Python/FastAPI, Node, Laravel)

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PROFESSIONAL SUMMARY

Applied AI Engineer with 7+ years across React/Next.js, Node/NestJS, Laravel, Python/FastAPI and Ionic/Capacitor, plus recent deep work in computer vision and local GPU inference. Designed and shipped Lumenweave, a visual-commerce room visualiser running segmentation, metric depth and FLUX.2 inpainting on a single RTX 3060. Also delivered a multi-app food platform, a 3D product viewer and a warehouse WMS. Leads frontend development and mentors developers.

TECHNICAL SKILLS

Computer Vision: SAM 3, SAM2, OWLv2, Depth Anything V2 (Metric Indoor Large), GeoCalib, SigLIP2, FLUX.2 [klein] inpainting, LaMa, Three.js
Local AI / Inference: GPU inference services, INT4 quantization (Nunchaku), Ollama, LM Studio, CUDA on RTX 3060 (12 GB)
Languages: TypeScript, JavaScript, Python, PHP, Rust
Frontend: React, Next.js, Vue, Vite, Tailwind CSS, shadcn/ui, TanStack Query, Zustand, React Hook Form + Zod
Backend & Data: FastAPI, Node.js, NestJS, Laravel, PostgreSQL, MariaDB, Prisma, Redis, Supabase, REST, GraphQL, WebSockets
Mobile: Ionic / Capacitor (Android, iOS) **DevOps & Testing:** Docker, GitHub Actions, Vitest, Jest, Cypress, Vercel, regression QA harnesses

PROFESSIONAL EXPERIENCE

Senior Full-Stack Developer | Victoria Plum **Jul 2024 – Present**
Remote — UK
Lead frontend delivery, mentoring developers and owning component architecture, code review and quality tooling.

Lumenweave — AI room visualiser *(Oct 2026)*
- Built a visual-commerce room visualiser: upload a room photo, remove furniture/mirrors/wall art, place catalogue products at real scale, with chained edits, undo and full-resolution PNG export as a React embed for retailer sites.
- Architected React/Vite + Three.js embed → Python FastAPI photo API → separate authenticated GPU inference service; integrated SAM 3, Depth Anything V2, GeoCalib, SigLIP2 and FLUX.2 [klein] INT4 inpainting (LaMa fallback) on a local RTX 3060 (12 GB).
- Warm object removal ~5–8 s; 0 px changed outside the effect mask; prompt prewarm cut cold encode ~13.6 s → ~3.3 s. Regression QA across 11 full-size rooms.

Global Foods — food-business platform *(Oct 2026)*
- Built a full-stack platform covering marketing site, customer portal, ops tools and EPOS tills (sales, collections, VAT), plus suppliers, statements, card payments and push notifications.
- Laravel/MariaDB with Reverb websockets and Docker queue workers; three Vue + Vite apps (customer, EPOS, ops); Ionic/Capacitor Android and iOS builds.

Giolindo 360 — Canvas 360° + 3D/AR product configurator
- Built Giolindo 360 for beds.co.uk: a 32-image Canvas 360° viewer and 3D/AR product configurator with admin model/material upload, original materials stripped on save, and runtime material switching.

Herd WMS
- Built a full-stack warehouse management system — NestJS, PostgreSQL, Redis, Prisma, WebSockets.

Frontend Developer | Next Order **Jul 2023 – Jul 2024**
Remote — Australia
- Built a React/TypeScript restaurant operations control panel (menus, vouchers, inventory, staff reports, analytics) with REST and TanStack Query data workflows.

Frontend Developer | Emumba **Sep 2020 – Jun 2023**
Islamabad, Pakistan
- Shipped fintech, fleet-tracking, analytics and crypto-tax interfaces in React/Next.js/TypeScript; Figma-to-code, complex forms, charts and reusable component architecture.

Frontend Developer | TheTekSol **Oct 2018 – Sep 2020**
Remote — Pakistan
- Built a production review-management system for COMSATS University thesis and final-year project evaluations (Laravel, JavaScript, Bootstrap).

SELECTED PROJECTS

YOLOPv2-Studio — Python/Flask studio for YOLOPv2 panoptic driving perception (object, lane and drivable-area detection) with image/webcam input and CUDA/MPS acceleration. github.com/usamakhangt4/YOLOPv2-Studio
Corpo Braindance — Local-first Next.js/React/TypeScript cyberpunk UI that generates and saves satirical corporate briefs in-browser (static export, Zustand, localStorage up to 20 briefs; no backend). Accessibility affordances (reduced motion/contrast/keyboard) with documented Vitest/smoke coverage. corpo-braindance.vercel.app
KittenTTS WebUI — Open-source (MIT) local web UI for the KittenTTS text-to-speech model. github.com/usamakhangt4/kitten-tts-webui · Live demo
git-auto-commit — Rust CLI for Ollama-based commit messages; local LLM stack on RTX 3060. github.com/usamakhangt4/git-auto-commit
viewer-360 — MIT Canvas Web Component 360° product viewer (zoom, high-res swap; React/Vue/Angular wrappers). github.com/usamakhangt4/viewer-360 · Live demo
shadcn-virtualized-tree — Typed React tree (headless + virtualized UI) for large hierarchies; MIT. github.com/usamakhangt4/shadcn-virtualized-tree · Live demo

EDUCATION

BS Software Engineering — COMSATS University Islamabad, Wah Campus **2016 – 2020**
Technical interests: computer vision, efficient on-device / local GPU inference, and end-to-end applied AI systems.